

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation Model Embeddings

When agents disagree, the satellite decides.

Every number an agent cites resolves to signed bytes, and re-reads to the pixel it came from.

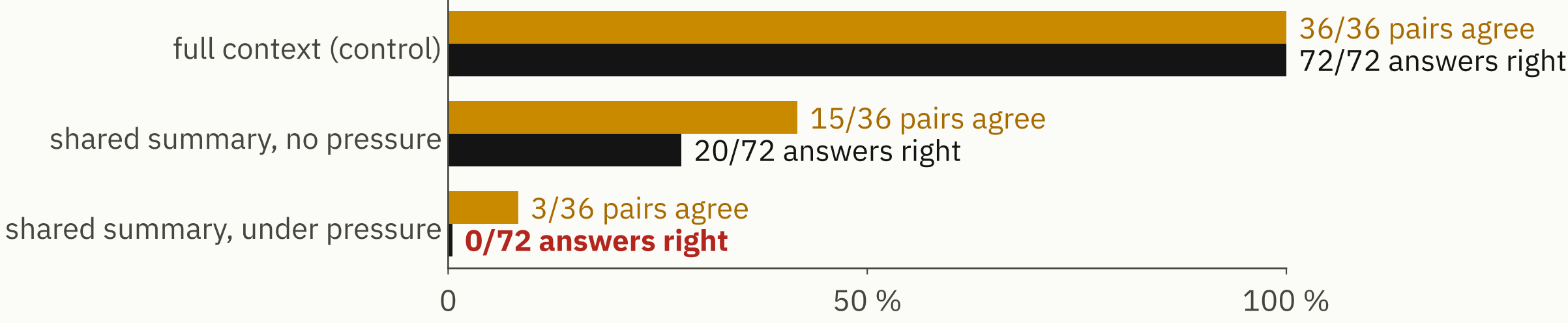
Jaya Kumari · Avijeet Singh · Vortex AI · avijeet@vortex.ai · doi:10.5281/zenodo.20706893 · github.com/Vortex-AI/emem (Apache-2.0) · emem.dev



THE PROBLEM

Agents that agree are not agents that are right.

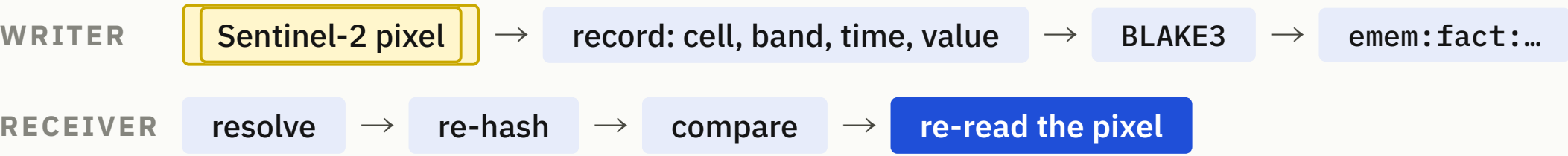
Each summary or handoff re-encodes a reading: the words survive, the referent does not. This is **referential drift**.



Pre-registered compaction study, Gemma-4-12B and Qwen2.5-7B on one host.

THE IDEA

Name each satellite observation by the hash of its bytes. Pass the name, not a sentence about it.

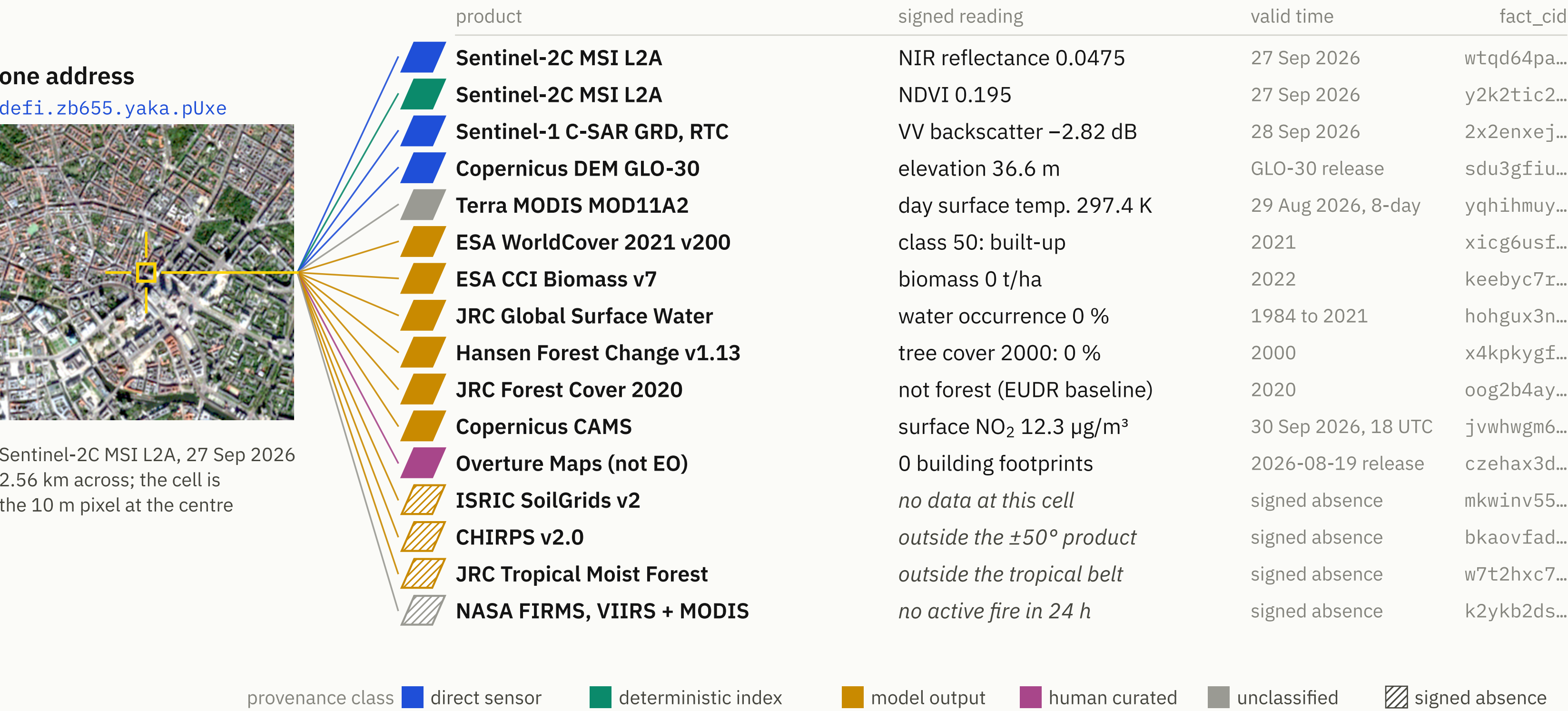


Names move
“the north field” drifts; a 64-bit cell id of about 10 m does not.

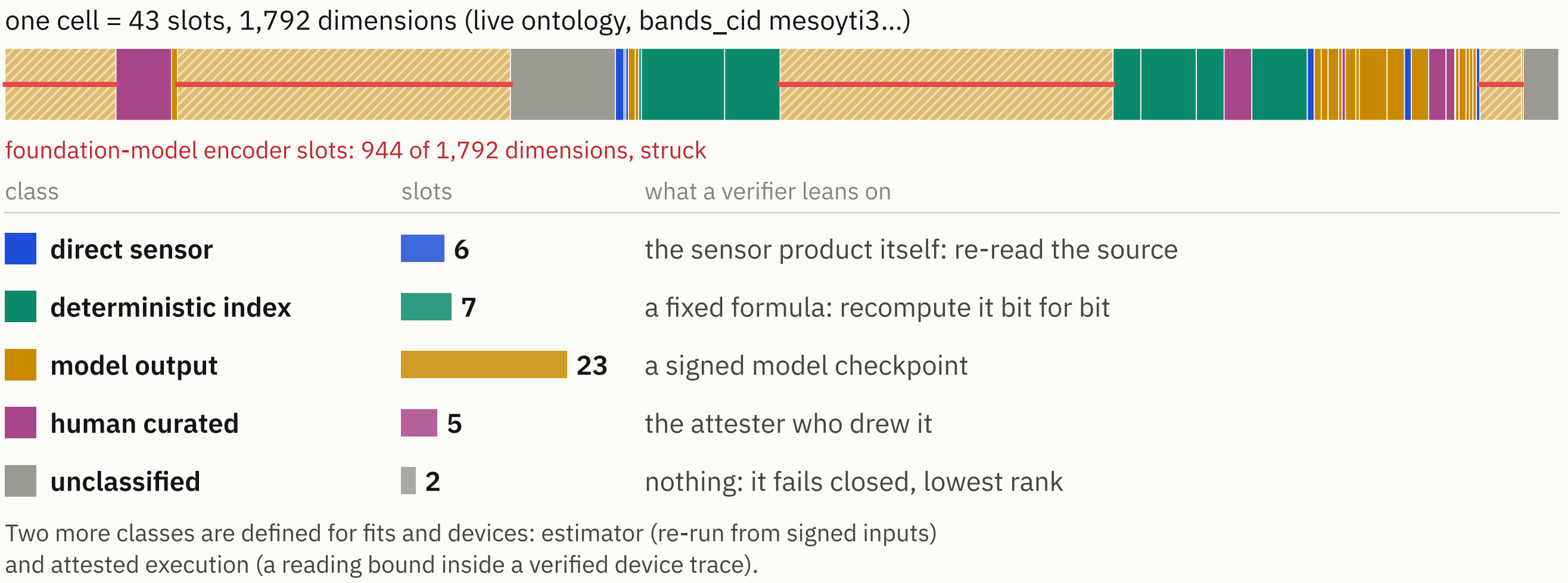
Values move
0.4709 becomes “about 0.47”; a fact_cid breaks if one bit changes.

1 One address, every satellite

One 10 m cell in central Berlin, read live on 30 Sep 2026: 15 products, 16 signed facts, 4 of them signed absences.



How emem encodes a memory

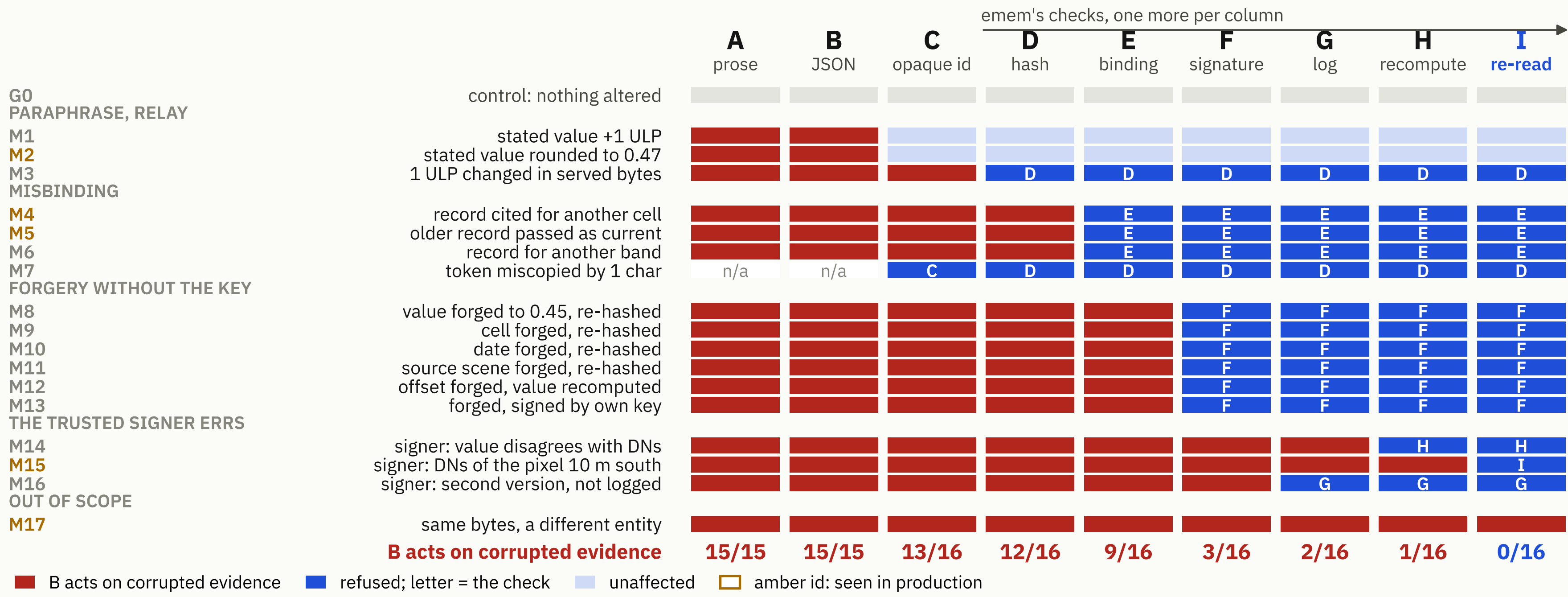


A fact is keyed by cell, band and valid time, written as canonical CBOR, named by BLAKE3 and signed with Ed25519. Its **provenance class** tells a verifier how far it can check.

Why the strike. An embedding is a model output: only its checkpoint vouches for it, and a new encoder moves every vector. emem still resolves every vector it signed, and builds on what can be re-read.

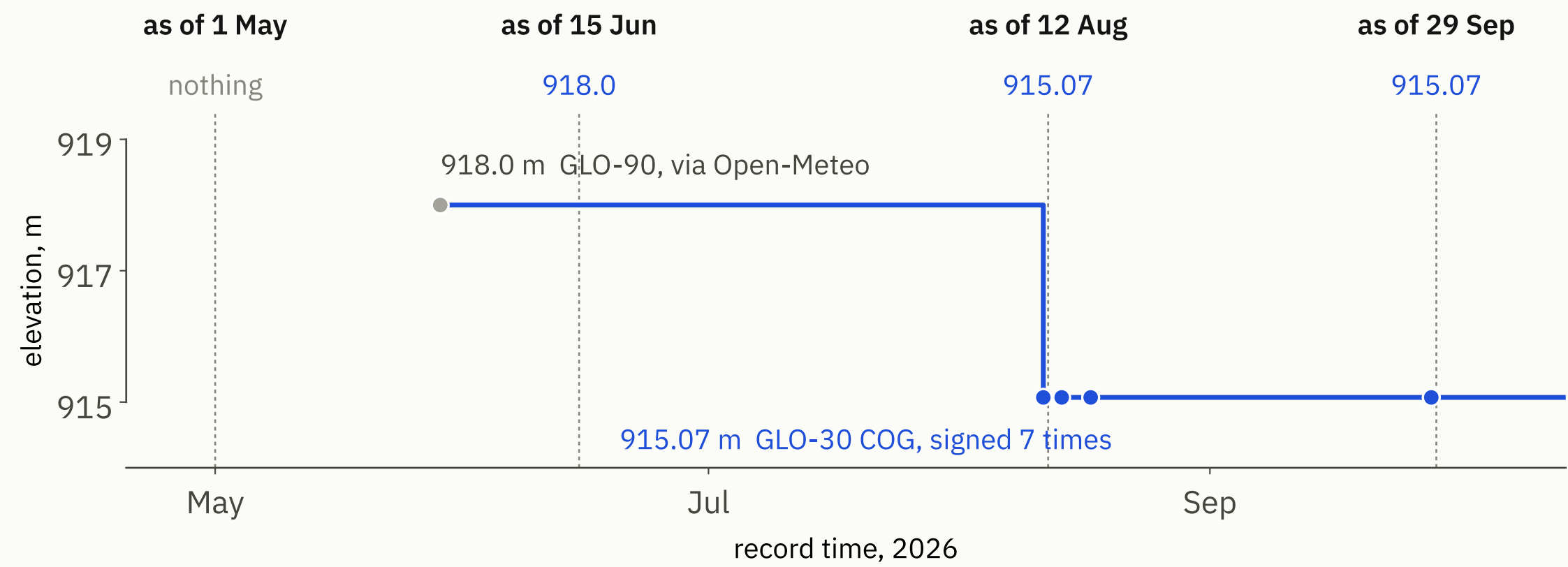
2 Seventeen ways to corrupt one reading

Agent B receives the Keylong NDVI record after one corruption. A deterministic verifier, no model in the loop, decides whether B acts.



M15 happened in production. The reader took the pixel 10 m south in 162 of 200 sampled records. Only re-reading the source caught it.

Memory on two clocks



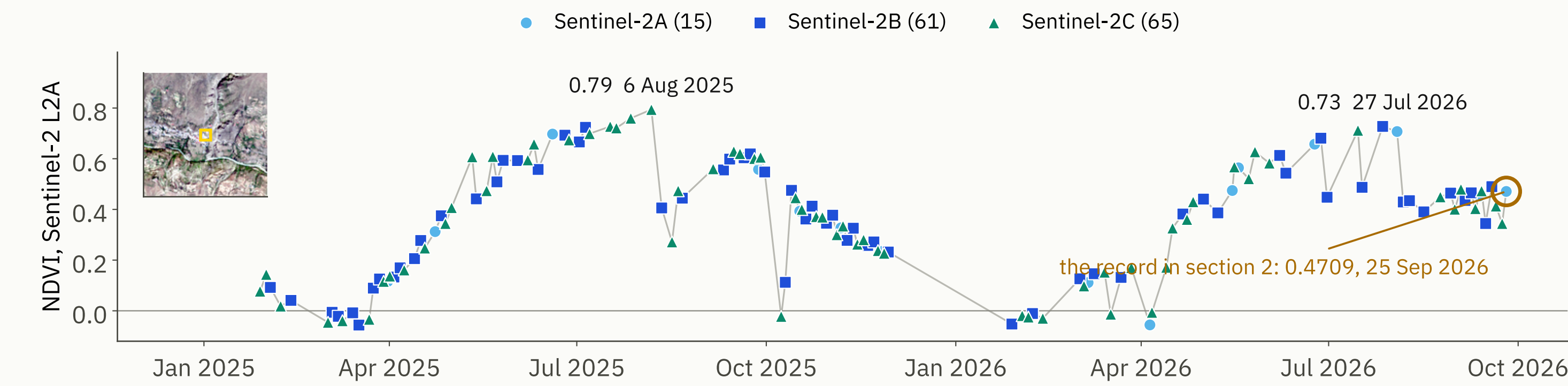
Bengaluru: 918.0 m from the 90 m Copernicus DEM in May, 915.07 m from the 30 m COG since 11 Aug. Nothing was overwritten; asked “as of 15 Jun”, emem still answers 918.0, and both answers verify.

Prose and JSON pass all 15 corruptions they can carry. With every check, none of the 16 in-scope cases gets through; offline, in under 2 ms.

3 Earth observation an agent can cite

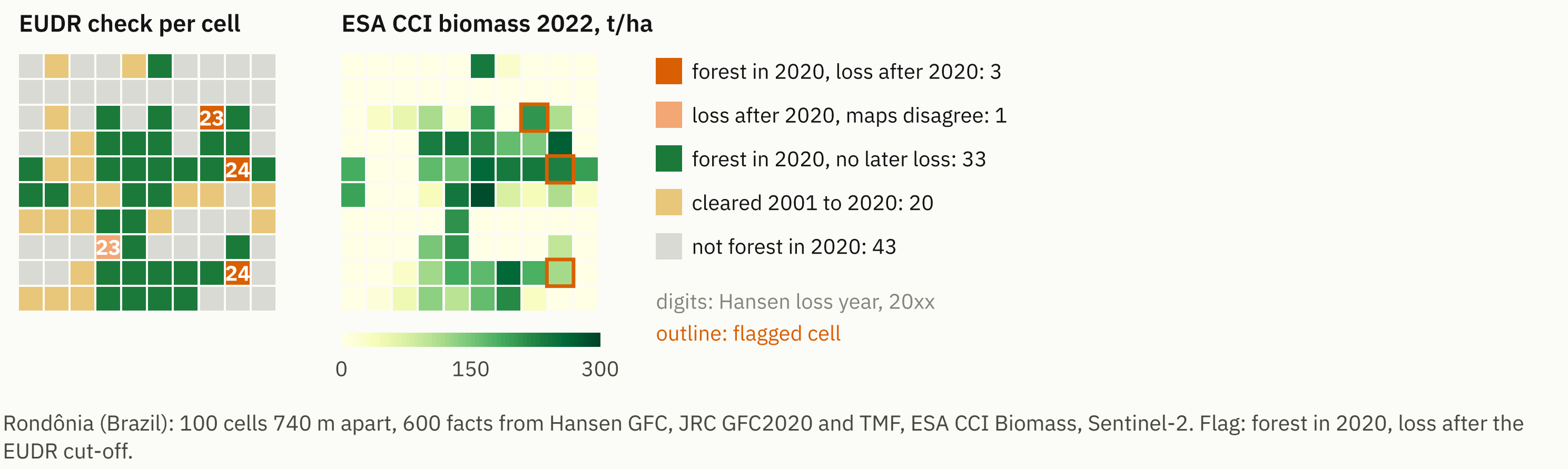
All 780 facts behind sections 1 and 3 were re-hashed, signature-checked and found in the log on 30 Sep 2026.

Two seasons, three satellites, one field



Keylong, Lahaul (India): 141 Sentinel-2 L2A NDVI facts at one 10 m cell, Jan 2025 to Sep 2026. Circled: the record corrupted in section 2.

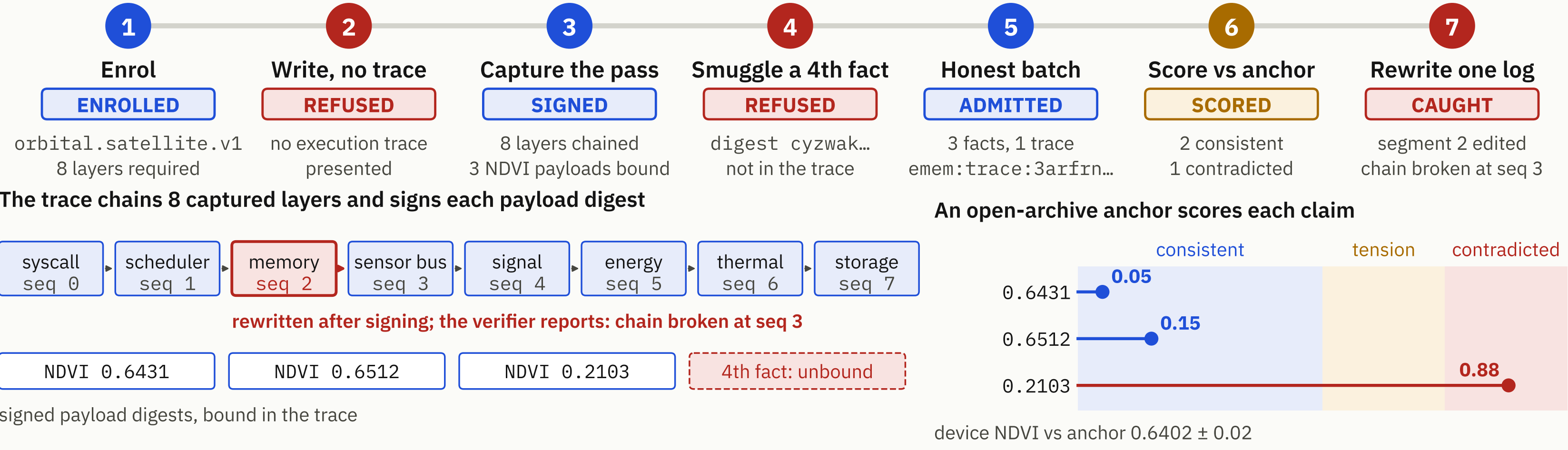
A deforestation check an auditor can re-run



Rondônia (Brazil): 100 cells 740 m apart, 600 facts from Hansen GFC, JRC GFC2020 and TMF, ESA CCI Biomass, Sentinel-2. Flag: forest in 2020, loss after the EUDR cut-off.

4 Satellites that prove what they ran

SAT-042, the reference spacecraft pass in emem's code, run on 30 Sep 2026.



Open archives enter emem by recomputation. A device cannot be recomputed, so its profile demands a **signed OS execution trace**: eight layers for orbital.satellite.v1, with every payload digest bound inside.

A write with no trace, and a fact the trace never emitted, are refused. Rewrite one log segment and the verifier names the broken link.

A trace proves what ran, not that the sensor was right. An open-archive anchor scores each claim.

What a verified token guarantees

INTEGRITY OF	HOW
the source bytes	guaranteed
the observation	guaranteed
the derivation	recomputable
the upstream file	named
the value's accuracy	inherited
the entity	out of scope

Objections, answered

A signature does not make a value true.

Agreed. It fixes which bytes were served. Truth is tested by re-reading the pixel; accuracy is the product's own validation.

Why not STAC, openEO or C2PA?

They find files, run processes, sign media. emem names the one observation an agent cites, so each of them can check it.

Is a satellite writing to emem today?

No spacecraft is enrolled. The trace verifier is live on emem.dev; SAT-042 exercises every refusal.

- Agents hand each other a signed observation, not a sentence about it.
- Each check stops a corruption the others miss; together, all 16.
- The memory never rewrites, so it answers what was known, and when.

Re-run every result. Data, scripts and claims map: research/repro/v12